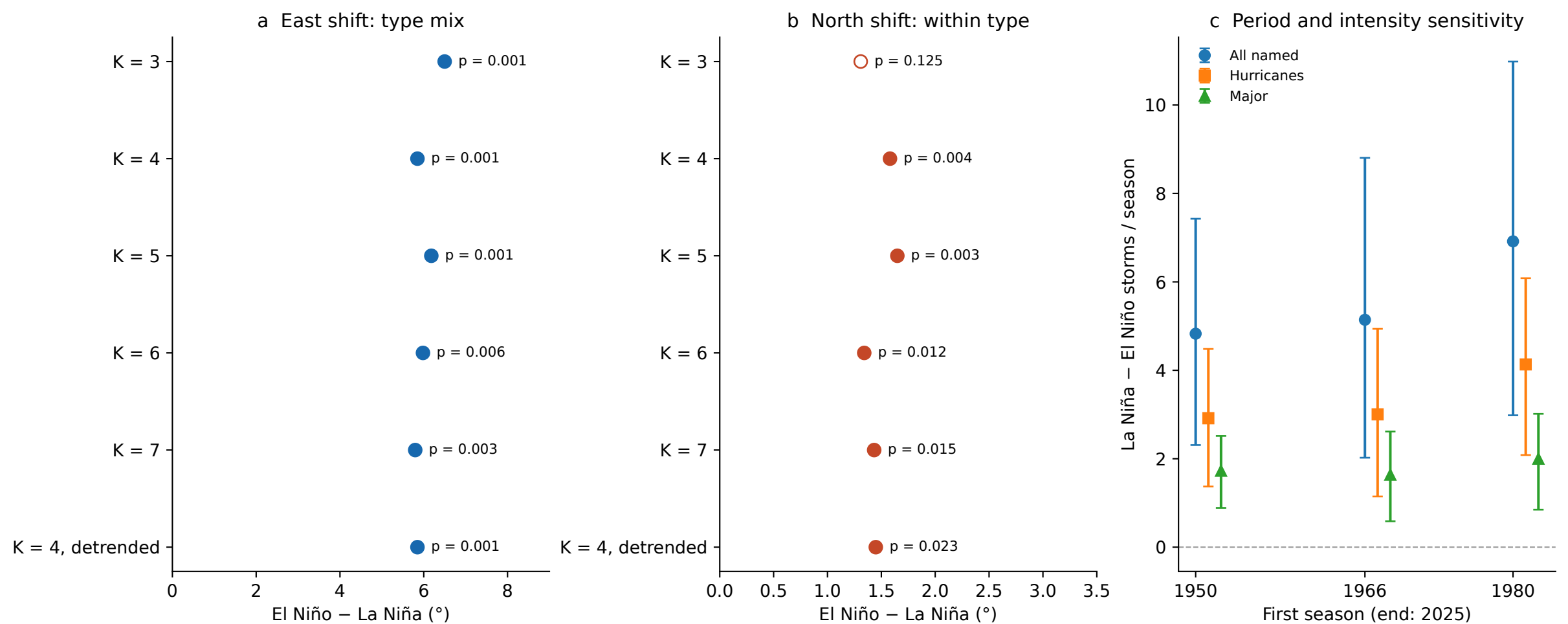
a East shift: type mix
K = 3
K = 4
K = 5
K = 6
K = 7
K = 4, detrended
p = 0.001
p = 0.001
p = 0.001
p = 0.006
p = 0.003
p = 0.001
0
2
4
6
8
El Niño − La Niña (°)
b North shift: within type
K = 3
K = 4
K = 5
K = 6
K = 7
K = 4, detrended
p = 0.125
p = 0.004
p = 0.003
p = 0.012
p = 0.015
p = 0.023
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
El Niño − La Niña (°)
c Period and intensity sensitivity
All named
Hurricanes
Major
La Niña − El Niño storms / season
0
2
4
6
8
10
1950
1966
1980
First season (end: 2025)